

Date: March 24, 2025

To: Financial Planners

From: Chloe, Stuart, Franklin

Subject: Update: Calculation Methodology for Old Age Security Benefits

A. Executive Summary

There are inconsistencies in advice on when clients should begin receiving Old Age Security (OAS) benefits. Our team built a Calculator, which incorporates inputs and formulas, that determines the optimal age for clients to start receiving payments.

OAS is a retirement pension for Canadian seniors, with amounts adjusted based on income and residency. To determine the optimal age for clients to begin receiving benefits, we analyzed mortality data from the Statistics Canada 2019-2023 life table and used a 3% interest rate to calculate the actuarial present value for each monthly deferral. The age with the highest present value is the optimal age for the client to begin receiving OAS payments.

The OAS Optimal Deferred Age Calculator uses inputs like gender, residency, and income. These factors significantly influence the optimal age for clients to begin receiving benefits. Key findings include:

- Gender: Males, with shorter life expectancies, receive payments earlier than females.
- Residency: Longer residency period increases the present value of payments.
- Income: Cannot receive payments in a given year if income from the previous year exceeds a certain threshold.
- Interest Rate: Higher rates decrease the present value of payments.

We recommend using the Calculator to provide personalized advice, helping clients maximize their OAS benefits. The default 3% interest rate should be applied unless clients specify otherwise.

B. Background Information

At the recent monthly meeting, Boss Louis identified inconsistencies in the recommendations provided to clients regarding when to begin receiving OAS benefits. To address this, Louis has tasked our actuarial team with conducting an analysis to provide clear guidance on the optimal age for clients to start receiving OAS payments.

The Old Age Security (OAS) program is a taxable monthly pension designed to assist Canadian seniors with basic living expenses. To qualify for OAS, individuals must meet certain criteria: [1]

- In Canada: Age 65+, Canadian citizen/legal resident, and 10+ years of residency since age 18.
- Outside Canada: Age 65+, Canadian citizen/legal resident before leaving, and 20+ years of residency since age 18.

To provide accurate, personalized advice for clients, we are using additional information on mortality rates and interest rates. This is important because the time value of money must be considered, and OAS payments continue until the client's death. We chose a reliable data source that provides up-to-date life expectancy data in Canada, broken down by gender. As for the interest rate, we selected a consistent rate that reflects real-world conditions. By balancing multiple data sources without overcomplicating calculations, we ensure our analysis stays accurate and practical.

The Statistics Canada 2019-2023 average life table provides detailed mortality data for Canadians, including death probabilities (q_x) between ages x and $x+1$. The table is segmented by sex and includes annual data from 2019 to 2023. [2] This table is a critical tool for actuarial analysis. We use the estimated life expectancy and mortality risks across different age groups to calculate actuarial present values.

The client's interest rate is assumed to be constant at 3%, equivalent to the average annual return on a 10-year zero-coupon bond. This assumption is based on the Government of Canada benchmark bond yields. [3] They are a reliable reference for long-term investment returns, ensuring our calculations are grounded in realistic and conservative estimates. This approach allows us to account for the time value of money while maintaining consistency in the Calculator.

This analysis and the Calculator serve as a tool for financial advisors to guide clients on when to begin receiving OAS payments.

C. Calculation Methodology

The OAS Optimal Deferred Age Calculator determines the most financially advantageous age to commence OAS benefits by calculating the maximum present value of expected future payments.

Key Assumptions

To ensure the generality of estimations, the following assumptions are applied:

Client Profile Factors:

1. Annual Net Income:

- 1.1. Clients will not receive any OAS benefits in the year, if last year's income reaches the threshold. [1]
- 1.2. The annual net income threshold is set in 2023: [1]

Below age 75	75 and above
\$142,609	\$148,179

- 1.3. We do not consider the OAS deduction from recovery tax.

2. Gender:

- 2.1. Gender group affects the mortality rates from Statistics Canada. [2]
- 2.2. Client's gender classifies the gender-specific life table to compute the corresponding actuarial present value.

3. Residency:

- 3.1. Eligibility requires at least ten years of residency in Canada after age 18, with benefits increasing proportionally up to a maximum of 40 years. [1]
- 3.2. We define the year of residency as the total number of positive integer years the client stayed in Canada at age 65.
- 3.3. If client is not eligible within age 75, the optimal solution is as soon as eligible.
- 3.4. We assume that after age 65, client will not leave the country with a period more than 6 months.

Actuarial Valuation Factors:**4. Interest Rate:**

- 4.1. Set at 3% by default, reflecting a 10-year government bond rate. [3] Further precision may be achieved by considering a client's individual financial status.

5. Inflation Rate:

- 5.1. The current maximum monthly payment at age 65 is set \$727.67. [1]
- 5.2. OAS payments are already indexed to Consumer Price Index (CPI), so inflation is excluded from the calculation. [1]
- 5.3. From age 75 onwards, client will gain 10% from the original payment. [1]

6. Mortality Rates:

- 6.1. We derived the mortality rate for each integer age with five-year national average from Statistics Canada. [2]
- 6.2. We assume clients have average health conditions and mortality risk as per Statistics Canada's life tables. [2]
- 6.3. We assume a uniform distribution of death (UDD) over every integer age, since we do not have actual monthly mortality rates while deferrals and payments are accrued monthly.
- 6.4. We do not need to consider clients with a current age greater than 75, otherwise the optimal present value is just the perpetuity at age 75.
- 6.5. We consider clients' mortality rate at age 110 and beyond is 100%, as per Statistics Canada's life tables. [2]

By incorporating these factors, we can apply the following actuarial formulas to calculate the actuarial present values for each monthly deferral.

Actuarial Formulas:

1. For each deferred month $u = 0, 1, \dots, 120$, compute:

$$\text{Total APV}(u) = \text{APV}(\text{all possible } u - \text{deferred payments before age 75}) \\ + \text{APV}(u - \text{deferred perpetuity starting at age 75})$$

2. The **Actuarial Present Value** for all possible u -deferred payments **before age 75**:

$$\text{APV} = \sum_{k=0}^{119} \left[\text{Payment}_u \times \frac{\text{survival probability to age } (65 + k \text{ months}) \text{ given at age 65}}{(1 + \text{interest rate})^{\frac{k}{12}}} \right]$$

- a. The survival probability from age 65 to age $65 + k$ months requires UDD assumption, see Appendix F2 for the formula derivations.

3. There are two formulas for the u -deferred payments,

- a. **Deferral-dominant** (Always dominant when Residency ≥ 15):

$$\text{Payment}_u^d = \$727.67 \times \left\{ \frac{\min([Residency], 40)}{40} + [1 + 7.2\%(\min(\frac{u}{12}, 5))] \right\}$$

- b. **Residency-dominant** (Always dominant when Residency ≤ 13):

$$\text{Payment}_u^r = \$727.67 \times \left\{ \frac{\min([Residency + \min(\frac{u}{12}, 5)], 40)}{40} + [1 + 7.2\%(\min(\frac{u}{12}, 5) - [\min(\frac{u}{12}, 5)])] \right\}$$

- c. Therefore, the actual **u -deferred payment** is:

$$\text{Payment}_u = \begin{cases} \max(\text{Payment}_u^d, \text{Payment}_u^r), & \text{if Residency} \geq 10 \text{ and Income} < \$142,609 \\ 0, & \text{otherwise.} \end{cases}$$

4. **u -deferred Perpetuity** starting at age 75:

$$\text{APV} = \sum_{k=120}^{540} \left[\text{Payment}_{u,k}^{75} \times \frac{\text{survival probability to age } (65 + k \text{ months}) \text{ given at age 65}}{(1 + \text{interest rate})^{\frac{k}{12}}} \right]$$

$$\text{where } \text{Payment}_{u,k}^{75} = 1.1 \times \text{Payment}_{u,k}$$

- a. By Assumption 6.5, the perpetuity only adds up to age 110, that is

$$k = (110 - 65) \times 12 = 540$$

5. The Maximum APV(u) among all $u = 0, 1, \dots, 120$ will yield the optimal deferral age $65 + \frac{u^*}{12}$:

$$\text{Total APV}(u^*) = \max_{u=0,1,\dots,120} \text{Total APV}(u)$$

D. Data Analysis:**1. Gender**

<u>Client's Input:</u>	By Gender		
Income	\$ 100,000	\$ 100,000	\$ 100,000
Gender	Female	Unspecified	Male
Residency	40	40	40
Interest Rate	3.0%	3.0%	3.0%
<u>Optimal solution:</u>			
Max PV	\$ 189,656	\$ 180,447	\$ 170,169
Deferral Age	70	69 9/12	69
<u>Monthly Payment:</u>			
Before age 75	\$ 989.63	\$ 976.53	\$ 937.24
75 and above	\$ 1,088.59	\$ 1,074.19	\$ 1,030.96

Male has lower survival rate than female, therefore early payments are emphasized, leading to the deferral one year earlier than female client.

2. Interest Rate

<u>Client's Input:</u>	By Interest Rate		
Income	\$ 100,000	\$ 100,000	\$ 100,000
Gender	Unspecified	Unspecified	Unspecified
Residency	40	40	40
Interest Rate	1.0%	3.0%	5.0%
<u>Optimal solution:</u>			
Max PV	\$ 209,498	\$ 180,447	\$ 157,561
Deferral Age	70	69 9/12	68 10/12
<u>Monthly Payment:</u>			
Before age 75	\$ 989.63	\$ 976.53	\$ 928.51
75 and above	\$ 1,088.59	\$ 1,074.19	\$ 1,021.36

Early payments are emphasized with higher interest rates, leading to the deferral at 5% interest rate 11 months earlier than one at 3%.

3. Years of Residency

<u>Client's Input:</u>	By Residency		
Income	\$ 100,000	\$ 100,000	\$ 100,000
Gender	Unspecified	Unspecified	Unspecified
Residency	16	15	14
Interest Rate	3.0%	3.0%	3.0%
<u>Optimal solution:</u>			
Max PV	\$ 72,179	\$ 67,668	\$ 63,927
Deferral Age	69 9/12	69 9/12	69 11/12
<u>Monthly Payment:</u>			
Before age 75	\$ 390.61	\$ 366.20	\$ 349.06
75 and above	\$ 429.67	\$ 402.82	\$ 383.97

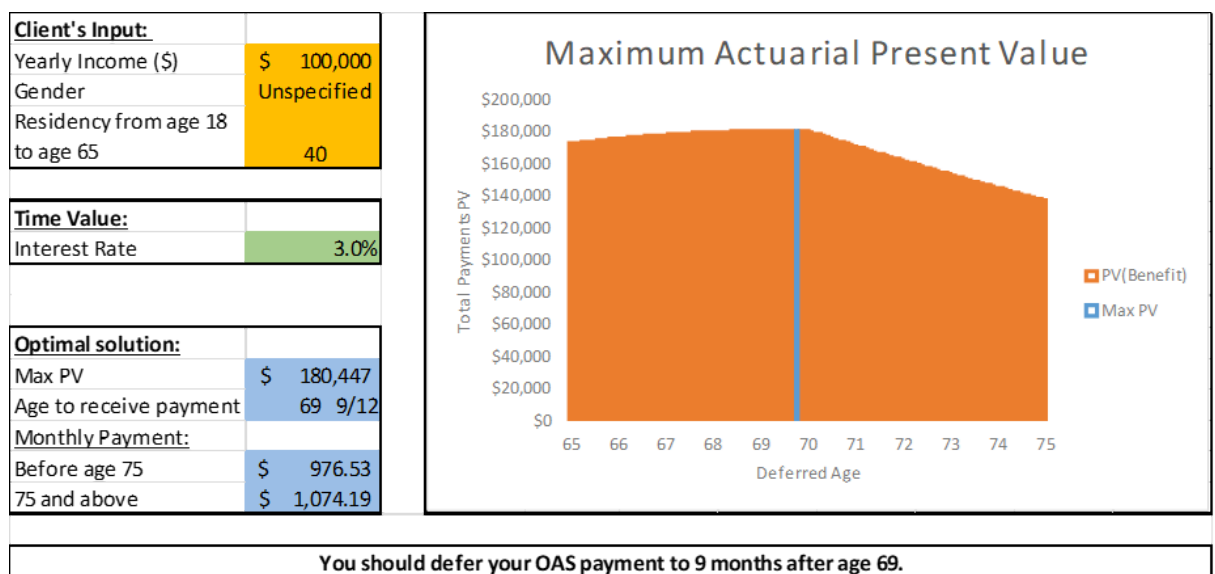
With the year of residency at age 65 less than 15 years, the change in payment formula will start affecting the optimal deferral age.

E. Recommendations

We recommend the use of the Calculator to determine the optimal age for clients to begin receiving OAS payments. This will require knowledge of the client's income, gender, and the number of years they have lived in Canada. The use of the Calculator will ensure that our clients receive consistent, accurate recommendations from our financial planners. Furthermore, we recommend that the input for the interest rate remains at 3%, unless the client specifies otherwise.

F. Appendix

- This is the dashboard of the optimal OAS deferral calculator:



- Monthly survival probability by UDD Assumption [4]:

$$\frac{k}{12} p_{65} = \text{survival probability to age } (65 + k \text{ months}) \text{ given at age } 65$$

$$= \left\lfloor \frac{k}{12} \right\rfloor p_{65} \left(\frac{k}{12} - \left\lfloor \frac{k}{12} \right\rfloor \right) p_{65 + \left\lfloor \frac{k}{12} \right\rfloor}$$

$$\stackrel{UDD}{=} \left\lfloor \frac{k}{12} \right\rfloor p_{65} \left[1 - \left(\frac{k}{12} - \left\lfloor \frac{k}{12} \right\rfloor \right) \times q_{65 + \left\lfloor \frac{k}{12} \right\rfloor} \right]$$

Both integer year probabilities $\left\lfloor \frac{k}{12} \right\rfloor p_{65}$ & $q_{65 + \left\lfloor \frac{k}{12} \right\rfloor}$ can be referred to the Life Table.

G. References

[1]	Canada.ca, “Old Age Security payment amount,” [Online]. Available: https://www.canada.ca/en/services/benefits/publicpensions/old-age-security/payments.html . [Accessed March 2025].
[2]	Statistics Canada, “Life expectancy and other elements of the complete life table, single-year estimates, Canada, all provinces except Prince Edward Island,” [Online]. Available: https://www150.statcan.gc.ca/t1/tbl1/en/tv.action?pid=1310083701 . [Accessed March 2025].
[3]	Bank of Canada, “Selected bond yields,” [Online]. Available: https://www.bankofcanada.ca/rates/interest-rates/canadian-bonds/ . [Accessed March 2025].
[4]	Dickson, D. C. M., Hardy, M. R., & Waters, H. R., Actuarial Mathematics for Life Contingent risks. Third Edition, Cambridge, 2019.